

Sport Coaching Tool

Database Schema

Current relational schema for authentication, teams, athletes, events, competitions, matches, statistics and standings.

Database	PostgreSQL (Neon)
ORM / migrations	Drizzle ORM / drizzle-kit
Total tables	12
Authentication tables	4
Application tables	8
Enums	5

Scope note. This document reflects the current repository schema at the time of analysis. It separates Better Auth infrastructure from Sport Coaching Tool application data.

1. Schema Overview

The database is centered around **users** and **teams**. Better Auth manages identity, sessions, authentication accounts and verification records. Application tables then model team membership, athletes, scheduled events, competitions, match details, per-athlete match statistics and competition standings.

A user may belong to multiple teams through the **team_members** junction table. Each team owns its athletes, events and competitions. Match-specific data extends an event through a one-to-one **matches** record. Athlete statistics are captured per match through **athlete_match_stats**.

Entity Relationship Diagram

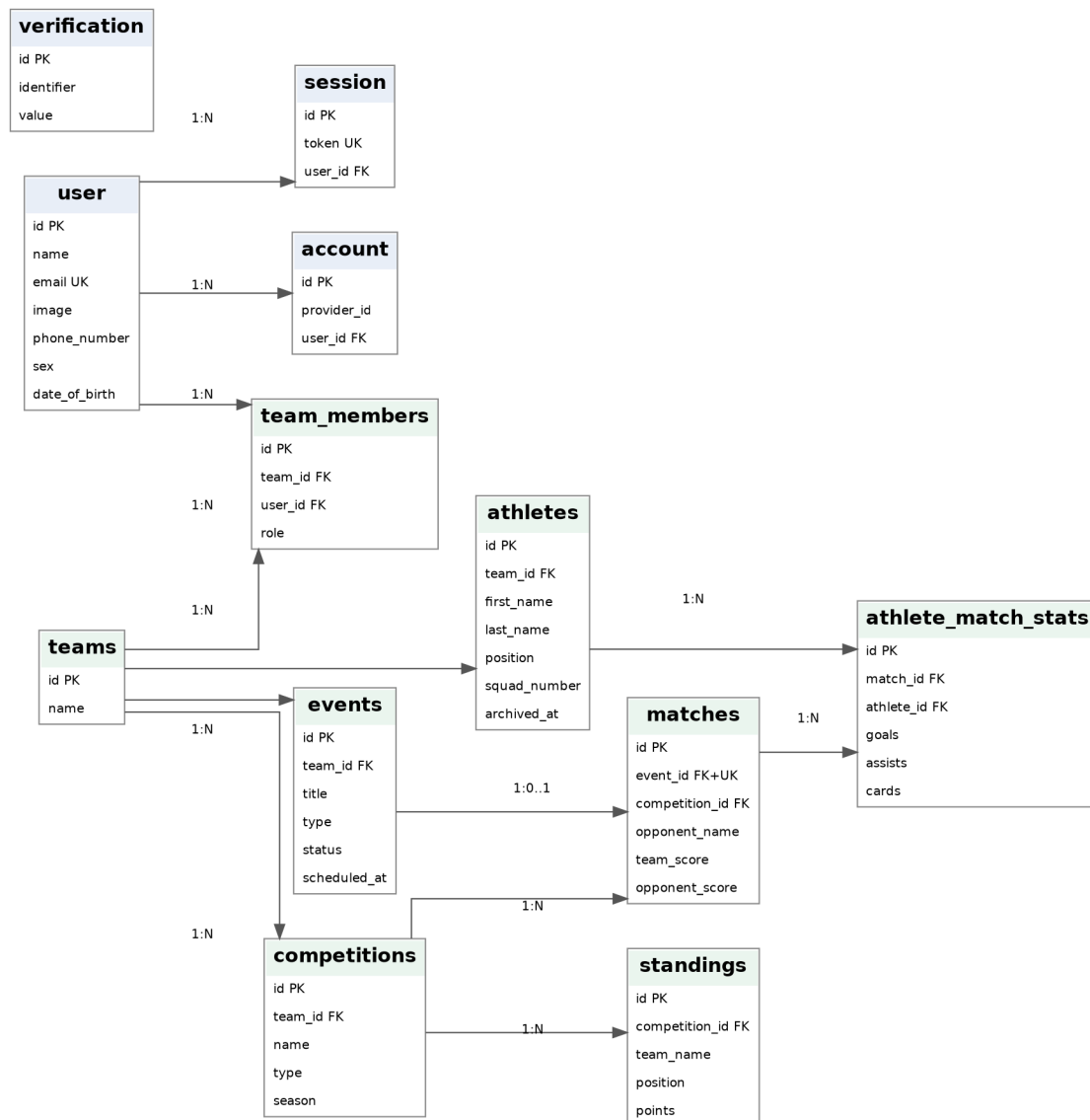


Diagram labels show primary keys (PK), foreign keys (FK), unique keys (UK), and high-level relationship cardinalities.

2. Better Auth Tables

These tables support authentication and identity. The **user** table is extended by the application with profile fields for phone number, sex and date of birth.

user

Purpose: Registered users and core identity data, extended with profile information.

Column	Type	Key	Nullable	Description
id	text	PK	No	Better Auth user identifier
name	text	-	No	Display name
email	text	Unique	No	Login identifier
email_verified	boolean	-	No	Email verification status
image	text	-	Yes	Avatar URL
phone_number	text	-	Yes	Profile phone number
sex	sex enum	-	Yes	male / female / prefer_not_to_say
date_of_birth	date	-	Yes	Profile date of birth
created_at	timestamptz	-	No	Created timestamp
updated_at	timestamptz	-	No	Updated timestamp

session

Purpose: Active authentication sessions associated with a user.

Column	Type	Key	Nullable	Description
id	text	PK	No	Session identifier
expires_at	timestamptz	-	No	Session expiry
token	text	Unique	No	Session token
ip_address	text	-	Yes	Client IP
user_agent	text	-	Yes	Browser user agent
user_id	text	FK -> user.id	No	Session owner

account

Purpose: Credential and OAuth account information associated with a user.

Column	Type	Key	Nullable	Description
id	text	PK	No	Account identifier
account_id	text	-	No	External/provider account id
provider_id	text	-	No	credential / google
user_id	text	FK -> user.id	No	Account owner
access_token	text	-	Yes	OAuth token
refresh_token	text	-	Yes	OAuth refresh token
password	text	-	Yes	Password hash for credential provider

verification

Purpose: One-time verification records such as email verification tokens.

Column	Type	Key	Nullable	Description
id	text	PK	No	Verification record id
identifier	text	-	No	Usually the email address
value	text	-	No	Verification token/hash
expires_at	timestampz	-	No	Expiry time

3. Sport Coaching Tool Application Tables

These tables represent the operational domain of the coaching platform.

teams

Purpose: A football team and the central organizational entity in the application.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Team id
name	text	-	No	Team name
created_at	timestamptz	-	No	Created timestamp
updated_at	timestamptz	-	No	Updated timestamp

team_members

Purpose: Junction table linking users to teams and assigning coach/assistant roles.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Membership row id
team_id	uuid	FK -> teams.id	No	Team
user_id	text	FK -> user.id	No	User
role	team_role enum	-	No	coach / assistant

athletes

Purpose: Players belonging to a team, with soft archive support.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Athlete id
team_id	uuid	FK -> teams.id	No	Owning team
first_name	text	-	No	First name
last_name	text	-	No	Last name
date_of_birth	date	-	Yes	Date of birth
position	text	-	Yes	Playing position
squad_number	integer	-	Yes	Jersey number
archived_at	timestamptz	-	Yes	Soft-delete/archive timestamp

events

Purpose: Scheduled team activities such as matches, training sessions and meetings.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Event id
team_id	uuid	FK -> teams.id	No	Owning team
title	text	-	No	Event title
type	event_type enum	-	No	match / training / meeting
status	event_status enum	-	No	scheduled / cancelled / completed
scheduled_at	timestamptz	-	No	Start time
location	text	-	No	Venue/address
notes	text	-	Yes	Optional notes

competitions

Purpose: Leagues, cups or friendlies associated with a team and season.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Competition id
team_id	uuid	FK -> teams.id	No	Owning team
name	text	-	No	Competition name
type	competition_type enum	-	No	league / cup / friendly
season	text	-	Yes	Season label

matches

Purpose: Extended match data linked one-to-one with an event.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Match id
event_id	uuid	FK + Unique	No	1:1 link to event
competition_id	uuid	FK	Yes	Optional competition
opponent_name	text	-	No	Opponent
is_home	boolean	-	No	Home/away flag
team_score	integer	-	No	Own score
opponent_score	integer	-	No	Opponent score

athlete_match_stats

Purpose: Per-athlete statistics for a specific match.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Stat row id
match_id	uuid	FK -> matches.id	No	Match
athlete_id	uuid	FK -> athletes.id	No	Athlete
started	boolean	-	No	Started match
minutes_played	integer	-	Yes	Minutes played
goals	integer	-	No	Goals
assists	integer	-	No	Assists
yellow_cards	integer	-	No	Yellow cards
red_cards	integer	-	No	Red cards

standings

Purpose: Competition standings rows entered for league/cup tables.

Column	Type	Key	Nullable	Description
id	uuid	PK	No	Standing row id
competition_id	uuid	FK -> competitions.id	No	Competition
team_name	text	-	No	Free-text team name
position	integer	-	No	Table position
played	integer	-	No	Played
won	integer	-	No	Won
drawn	integer	-	No	Drawn
lost	integer	-	No	Lost
goals_for	integer	-	No	Goals for
goals_against	integer	-	No	Goals against
points	integer	-	No	Points
is_own_team	boolean	-	No	Highlights own team

4. Relationships and Delete Behavior

The following relationships are enforced through foreign keys. Most child records use **ON DELETE CASCADE** so dependent data is removed when a parent is deleted. The competition link on matches is the exception: deleting a competition sets **matches.competition_id** to NULL so the match record is retained.

Parent	Child FK	Cardinality	On delete
user.id	session.user_id	1-to-many	CASCADE
user.id	account.user_id	1-to-many	CASCADE
user.id	team_members.user_id	1-to-many	CASCADE
teams.id	team_members.team_id	1-to-many	CASCADE
teams.id	athletes.team_id	1-to-many	CASCADE
teams.id	events.team_id	1-to-many	CASCADE
teams.id	competitions.team_id	1-to-many	CASCADE
events.id	matches.event_id	1-to-0..1	CASCADE
competitions.id	matches.competition_id	1-to-many	SET NULL
competitions.id	standings.competition_id	1-to-many	CASCADE
matches.id	athlete_match_stats.match_id	1-to-many	CASCADE
athletes.id	athlete_match_stats.athlete_id	1-to-many	CASCADE

Implicit many-to-many relationships: users and teams are many-to-many through **team_members**; matches and athletes are many-to-many through **athlete_match_stats**, which also stores performance data.

5. Enumerations

Enum	Allowed values	Used in
sex	male, female, prefer_not_to_say	user.sex
team_role	coach, assistant	team_members.role
event_type	match, training, meeting	events.type
event_status	scheduled, cancelled, completed	events.status
competition_type	league, cup, friendly	competitions.type

6. Design Notes

Athlete soft archive. The **athletes.archived_at** column is the only soft-delete field in the current schema. A non-null value indicates an archived athlete; restoring an athlete clears the timestamp.

Match extension. **matches.event_id** is unique, enforcing at most one match row per event. This allows common scheduling information to remain in **events** while match-specific score/opponent data is separated.

Standings. **standings.team_name** is stored as free text rather than a foreign key to **teams**. This supports recording opponents that are not registered as teams in the application.

Profile data. The Better Auth **user** table is extended with phone number, sex and date of birth. The existing **image** field is reused for avatar URLs.

Authentication. The schema supports email/password and Google OAuth through Better Auth. Session/account child rows cascade when a user is deleted.

Implementation note

Repository analysis identified an orphan migration file named **0002_absent_the_stranger.sql** that is not listed in the Drizzle migration journal and appears to duplicate a later migration. This should be reviewed by the development team before cleanup; it is not represented as part of the applied schema in this document.